MTRC

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References Book:

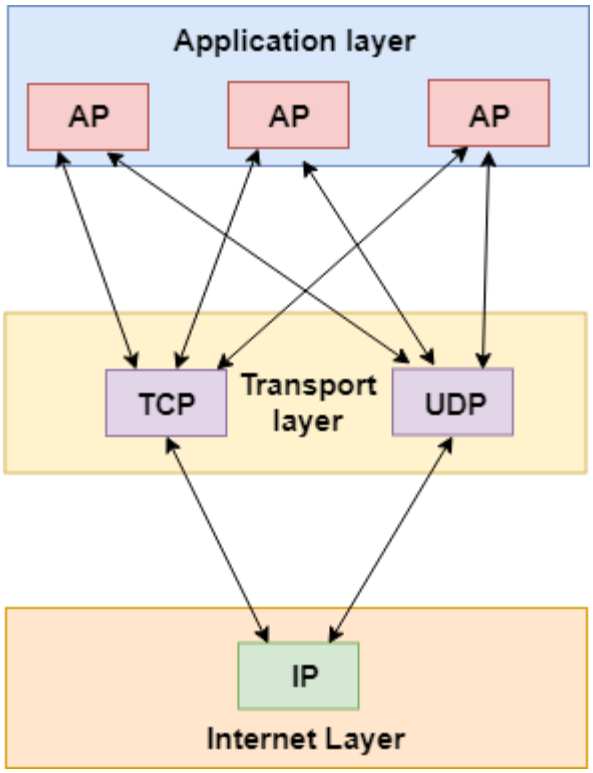
1. COMPUTER NETWORK,ANDREW S. TANENBAUM 5TH EDITION PEARSON PUBLICTION
2. COMPUTER NETWORKING JAMES F KUROSE PEARSON PUBLICATION
3. COMPUTER NETWORKING BEHROUZ A FOROUZAN TMH PUBLICATION

FACULTY HOD 17/10/2024 DEAN ACADEMICS

UNIT 4

Transport Layer

- The transport layer is a 4th layer from the top.
- The main role of the transport layer is to provide the communication services directly to the application processes running on different hosts.
- The transport layer provides a logical communication between application processes running on different hosts. Although the application processes on different hosts are not physically connected, application processes use the logical communication provided by the transport layer to send the messages to each other.
- The transport layer protocols are implemented in the end systems but not in the network routers.
- A computer network provides more than one protocol to the network applications. For example, TCP and UDP are two transport layer protocols that provide a different set of services to the network layer.
- All transport layer protocols provide multiplexing/demultiplexing service. It also provides other services such as reliable data transfer, bandwidth guarantees, and delay guarantees.
- Each of the applications in the application layer has the ability to send a message by using TCP or UDP. The application communicates by using either of these two protocols. Both TCP and UDP will then communicate with the internet protocol in the internet layer. The applications can read and write to the transport layer. Therefore, we can say that communication is a two-way process.

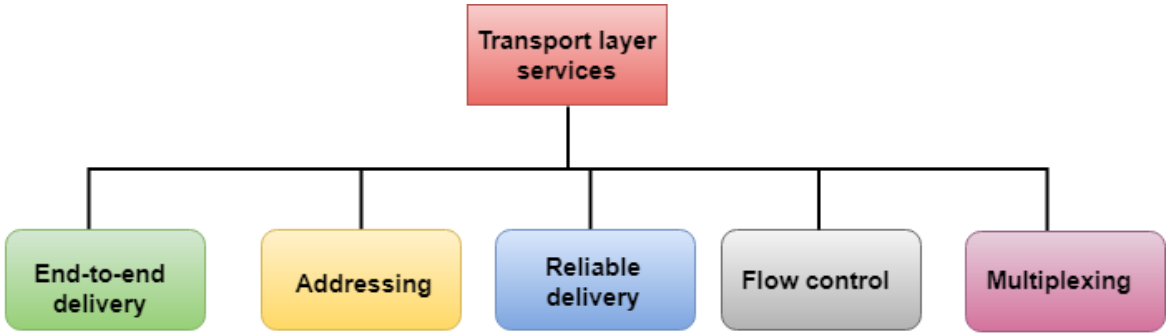


Services provided by the Transport Layer

The services provided by the transport layer are similar to those of the data link layer. The data link layer provides the services within a single network while the transport layer provides the services across an internetwork made up of many networks. The data link layer controls the physical layer while the transport layer controls all the lower layers.

The services provided by the transport layer protocols can be divided into five categories:

- End-to-end delivery
- Addressing
- Reliable delivery
- Flow control
- Multiplexing



End-to-end delivery:

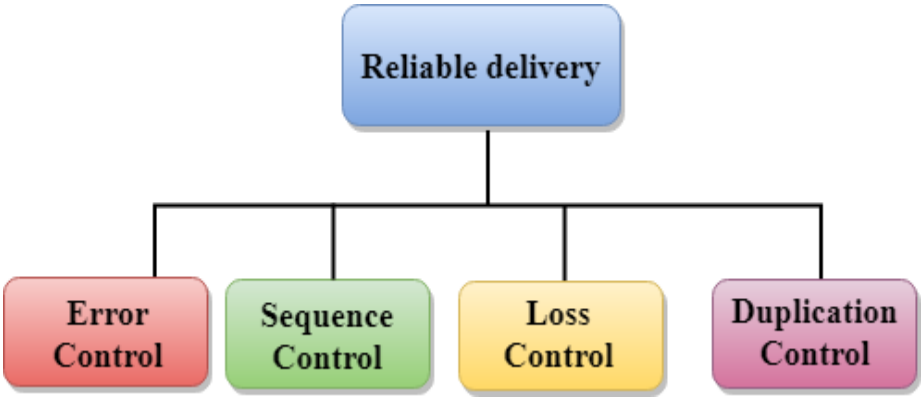
The transport layer transmits the entire message to the destination. Therefore, it ensures the end-to-end delivery of an entire message from a source to the destination.

Reliable delivery:

The transport layer provides reliability services by retransmitting the lost and damaged packets.

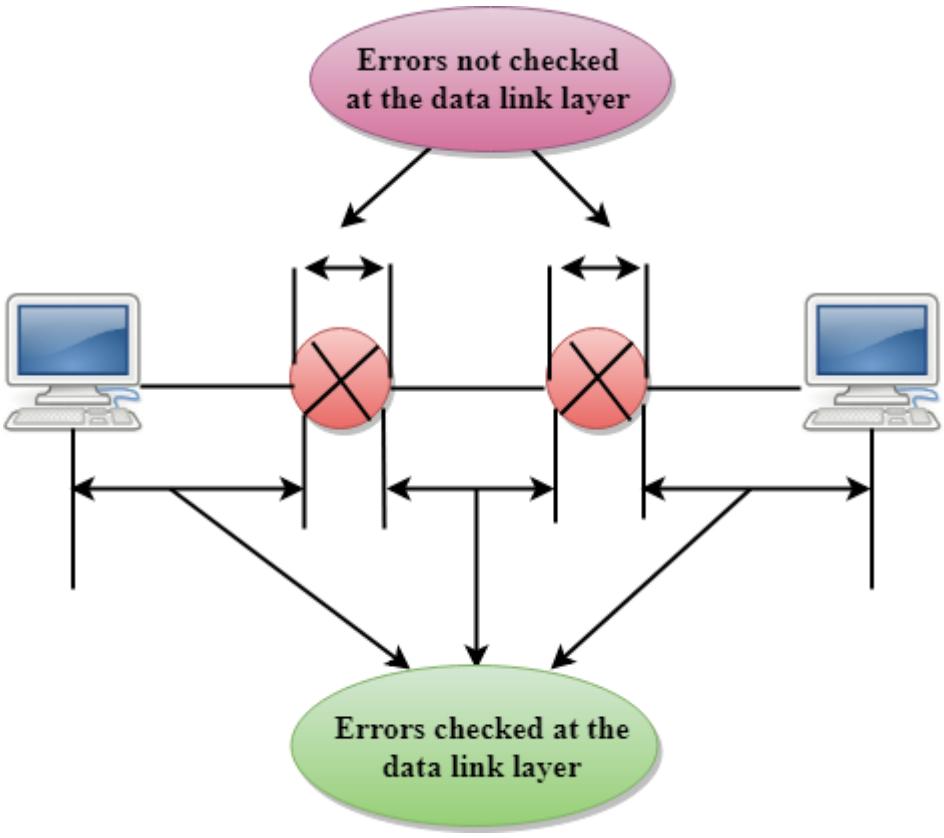
The reliable delivery has four aspects:

- Error control
- Sequence control
- Loss control
- Duplication control



Error Control

- The primary role of reliability is **Error Control**. In reality, no transmission will be 100 percent error-free delivery. Therefore, transport layer protocols are designed to provide error-free transmission.
- The data link layer also provides the error handling mechanism, but it ensures only node-to-node error-free delivery. However, node-to-node reliability does not ensure the end-to-end reliability.
- The data link layer checks for the error between each network. If an error is introduced inside one of the routers, then this error will not be caught by the data link layer. It only detects those errors that have been introduced between the beginning and end of the link. Therefore, the transport layer performs the checking for the errors end-to-end to ensure that the packet has arrived correctly.



Sequence Control

- The second aspect of the reliability is sequence control which is implemented at the transport layer.
- On the sending end, the transport layer is responsible for ensuring that the packets received from the upper layers can be used by the lower layers. On the receiving end, it ensures that the various segments of a transmission can be correctly reassembled.

Loss Control

Loss Control is a third aspect of reliability. The transport layer ensures that all the fragments of a transmission arrive at the destination, not some of them. On the sending end, all the fragments of transmission are given sequence numbers by a transport layer. These sequence numbers allow the receiver?s transport layer to identify the missing segment.

Duplication Control

Duplication Control is the fourth aspect of reliability. The transport layer guarantees that no duplicate data arrive at the destination. Sequence numbers are used to identify the lost packets; similarly, it allows the receiver to identify and discard duplicate segments.

Flow Control

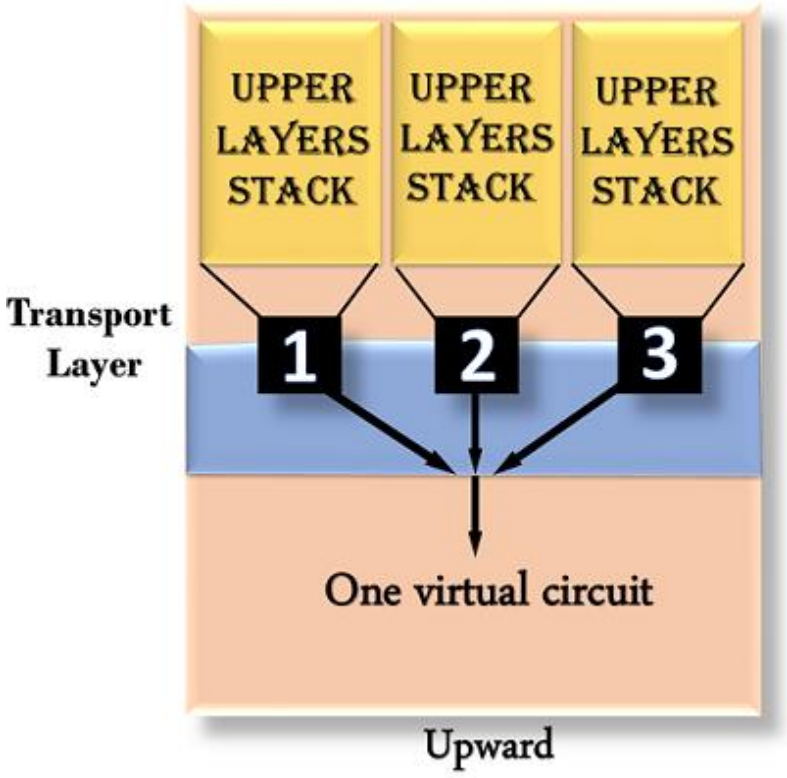
Flow control is used to prevent the sender from overwhelming the receiver. If the receiver is overloaded with too much data, then the receiver discards the packets and asking for the retransmission of packets. This increases network congestion and thus, reducing the system performance. The transport layer is responsible for flow control. It uses the sliding window protocol that makes the data transmission more efficient as well as it controls the flow of data so that the receiver does not become overwhelmed. Sliding window protocol is byte oriented rather than frame oriented.

Multiplexing

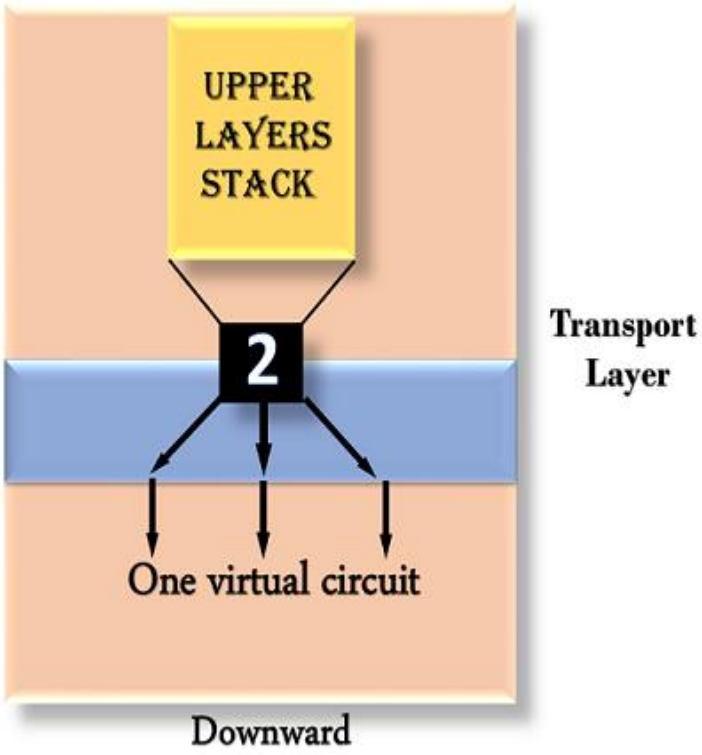
The transport layer uses the multiplexing to improve transmission efficiency.

Multiplexing can occur in two ways:

- **Upward multiplexing:** Upward multiplexing means multiple transport layer connections use the same network connection. To make more cost-effective, the transport layer sends several transmissions bound for the same destination along the same path; this is achieved through upward multiplexing.

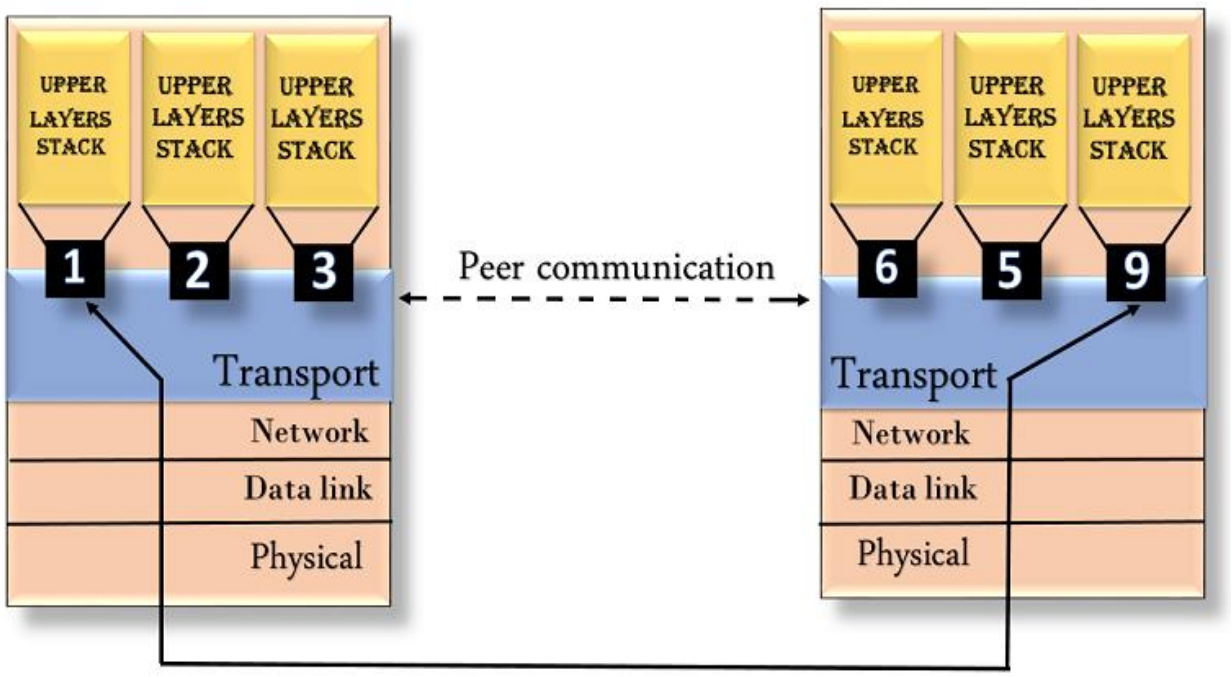


- **Downward multiplexing:** Downward multiplexing means one transport layer connection uses the multiple network connections. Downward multiplexing allows the transport layer to split a connection among several paths to improve the throughput. This type of multiplexing is used when networks have a low or slow capacity.



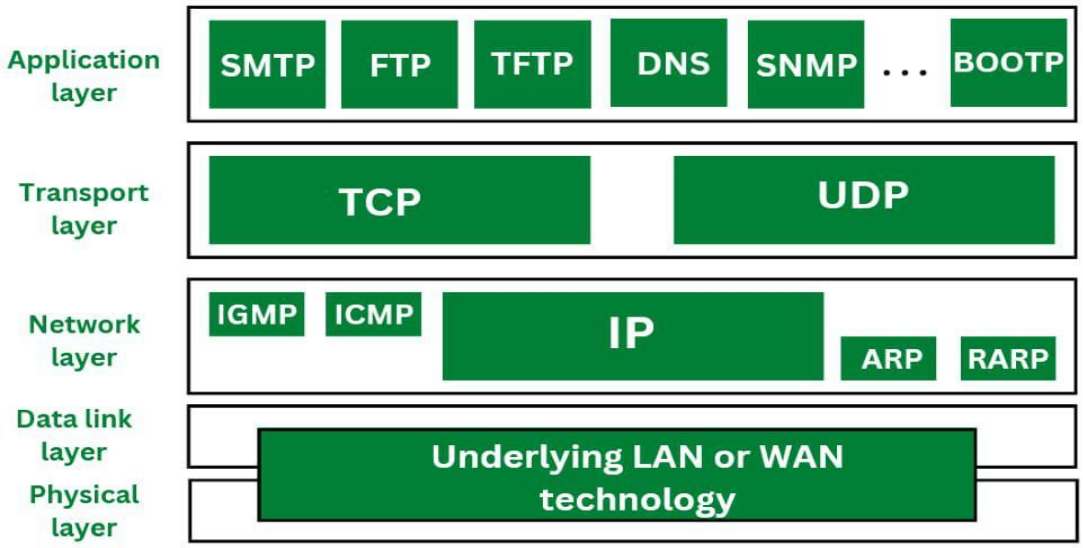
Addressing

- According to the layered model, the transport layer interacts with the functions of the session layer. Many protocols combine session, presentation, and application layer protocols into a single layer known as the application layer. In these cases, delivery to the session layer means the delivery to the application layer. Data generated by an application on one machine must be transmitted to the correct application on another machine. In this case, addressing is provided by the transport layer.
- The transport layer provides the user address which is specified as a station or port. The port variable represents a particular TS user of a specified station known as a Transport Service access point (TSAP). Each station has only one transport entity.
- The transport layer protocols need to know which upper-layer protocols are communicating.



TRANSPORT LAYER PROTOCOLS

The transport layer is the fourth layer in the OSI model and the second layer in the TCP/IP model. The transport layer provides with end to end connection between the source and the destination and reliable delivery of the services. Therefore transport layer is known as the end-to-end layer. The transport layer takes the services from its upward layer which is the application layer and provides it to the network layer. Segment is the unit of data encapsulation at the transport layer.



Functions of Transport Layer

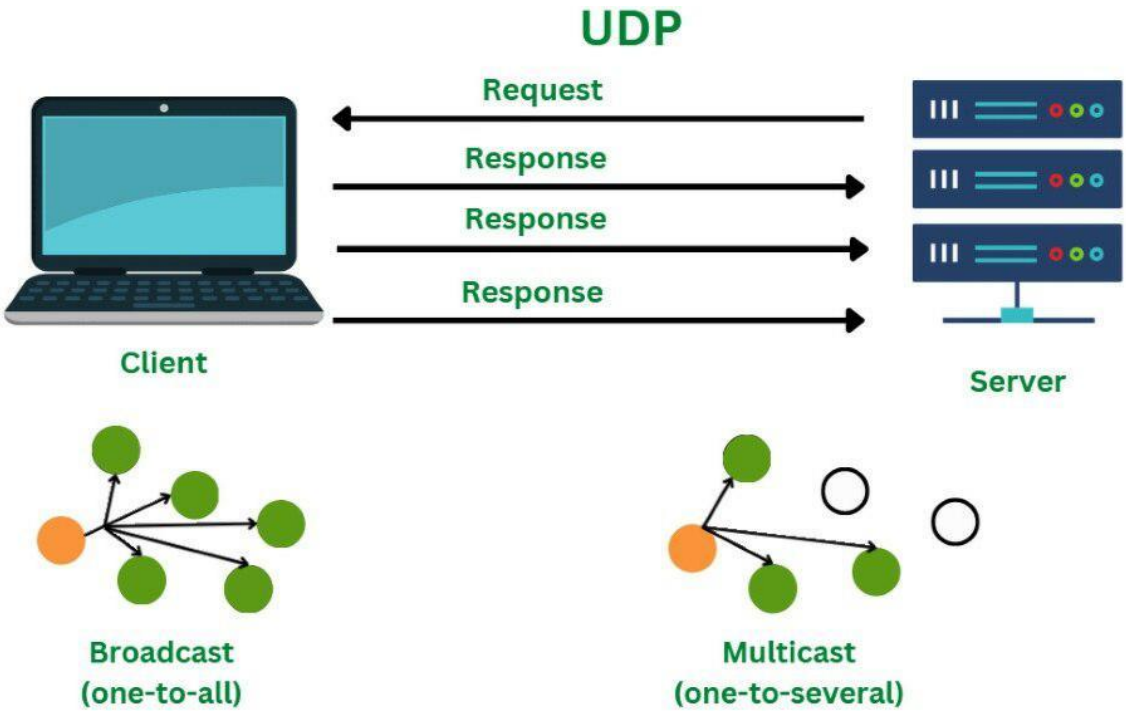
- Process to process delivery
- End-to-end connection between devices
- Multiplexing and Demultiplexing
- Data integrity and error Correction
- Congestion Control
- Flow Control

Transport Layer Protocols

The transport layer is represented majorly by TCP and UDP protocols. Today almost all operating systems support multiprocessing multi-user environments. This transport layer protocol provides connections to the individual ports. These ports are known as protocol ports. Transport layer protocols work above the IP protocols and deliver the data packets from IP serves to destination port and from the originating port to destination IP services. Below are the protocols used at the transport layer.

1. UDP

UDP stands for User Datagram Protocol. User Datagram Protocol provides a nonsequential transmission of data. It is a connectionless transport protocol. UDP protocol is used in applications where the speed and size of data transmitted is considered as more important than the security and reliability. User Datagram is defined as a packet produced by User Datagram Protocol. UDP protocol adds checksum error control, transport level addresses, and information of length to the data received from the layer above it. Services provided by User Datagram Protocol(UDP) are connectionless service, faster delivery of messages, checksum, and process-to-process communication.



Advantages of UDP

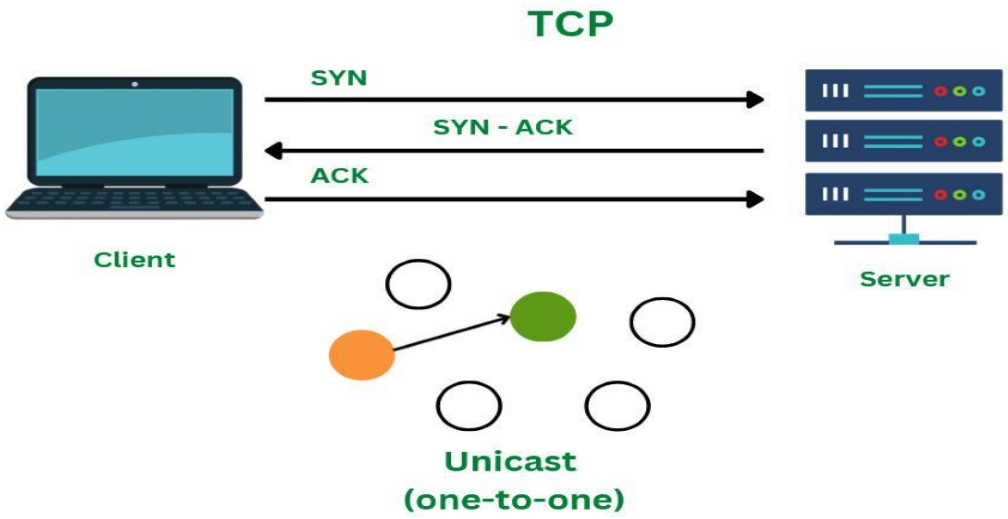
- UDP also provides multicast and broadcast transmission of data.
- UDP protocol is preferred more for small transactions such as DNS lookup.
- It is a connectionless protocol, therefore there is no compulsion to have a connection-oriented network.
- UDP provides fast delivery of messages.

Disadvantages of UDP

- In UDP protocol there is no guarantee that the packet is delivered.
- UDP protocol suffers from worse packet loss.
- UDP protocol has no congestion control mechanism.
- UDP protocol does not provide the sequential transmission of data.

2. TCP

TCP stands for Transmission Control Protocol. TCP protocol provides transport layer services to applications. TCP protocol is a connection-oriented protocol. A secured connection is being established between the sender and the receiver. For a generation of a secured connection, a virtual circuit is generated between the sender and the receiver. The data transmitted by TCP protocol is in the form of continuous byte streams. A unique sequence number is assigned to each byte. With the help of this unique number, a positive acknowledgment is received from receipt. If the acknowledgment is not received within a specific period the data is retransmitted to the specified destination.



Advantages of TCP

- TCP supports multiple routing protocols.
- TCP protocol operates independently of that of the operating system.
- TCP protocol provides the features of error control and flow control.
- TCP provides a connection-oriented protocol and provides the delivery of data.

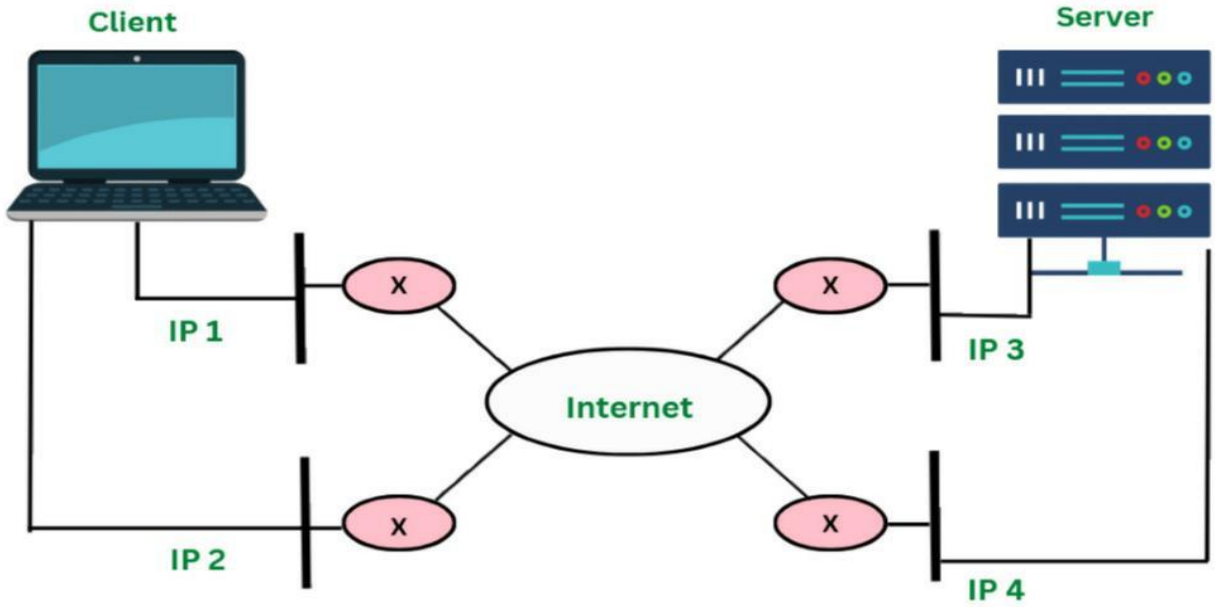
Disadvantages of TCP

- TCP protocol cannot be used for broadcast or multicast transmission.
- TCP protocol has no block boundaries.
- No clear separation is being offered by TCP protocol between its interface, services, and protocols.
- In TCP/IP replacement of protocol is difficult.

3. SCTP

SCTP stands for Stream Control Transmission Protocol. SCTP is a connection-oriented protocol. Stream Control Transmission Protocol transmits the data from sender to receiver in full duplex mode. SCTP is a unicast protocol that provides with point to point-to-point connection and uses different hosts for reaching the destination. SCTP protocol provides a simpler way to build a connection over a wireless network. SCTP protocol provides a reliable transmission of data. SCTP provides a reliable and easier telephone conversation over the internet. SCTP protocol supports the feature of multihoming ie. it can establish more than one connection path between the two points of communication and does not depend on the IP

layer. SCTP protocol also ensures security by not allowing the half-open connections.



Advantages of SCTP

- SCTP provides a full duplex connection. It can send and receive the data simultaneously.
- SCTP protocol possesses the properties of both TCP and UDP protocol.
- SCTP protocol does not depend on the IP layer.
- SCTP is a secure protocol.

Disadvantages of SCTP

- To handle multiple streams simultaneously the applications need to be modified accordingly.
- The transport stack on the node needs to be changed for the SCTP protocol.
- Modification is required in applications if SCTP is used instead of TCP or UDP protocol

TCP CONNECTION ESTABLISHMENT

TCP 3-Way Handshake Process

TCP is a connection-oriented protocol and every connection-oriented protocol needs to establish a connection in order to reserve resources at both the communicating ends.

Connection Establishment –

1. Sender starts the process with the following:

- **Sequence number (Seq=521):** contains the random initial sequence number generated at the sender side.
- **Syn flag (Syn=1):** request the receiver to synchronize its sequence number with the above-provided sequence number.
- **Maximum segment size (MSS=1460 B):** sender tells its maximum segment size, so that receiver sends datagram which won't require any fragmentation. MSS field is present inside **Option** field in TCP header.
- **Window size (window=14600 B):** sender tells about his buffer capacity in which he has to store messages from the receiver.

2. TCP is a full-duplex protocol so both sender and receiver require a window for receiving messages from one another.

- **Sequence number (Seq=2000):** contains the random initial sequence number generated at the receiver side.
- **Syn flag (Syn=1):** request the sender to synchronize its sequence number with the above-provided sequence number.
- **Maximum segment size (MSS=500 B):** receiver tells its maximum segment size, so that sender sends datagram which won't require any fragmentation. MSS field is present inside **Option** field in TCP header.
  Since $MSS_{receiver} < MSS_{sender}$, both parties agree for minimum MSS i.e., 500 B to avoid fragmentation of packets at both ends.

Therefore, receiver can send maximum of 14600/500 = 29 packets.

This is the receiver's sending window size.

- **Window size (window=10000 B):** receiver tells about his buffer capacity in which he has to store messages from the sender.

Therefore, sender can send a maximum of 10000/500 = 20 packets.

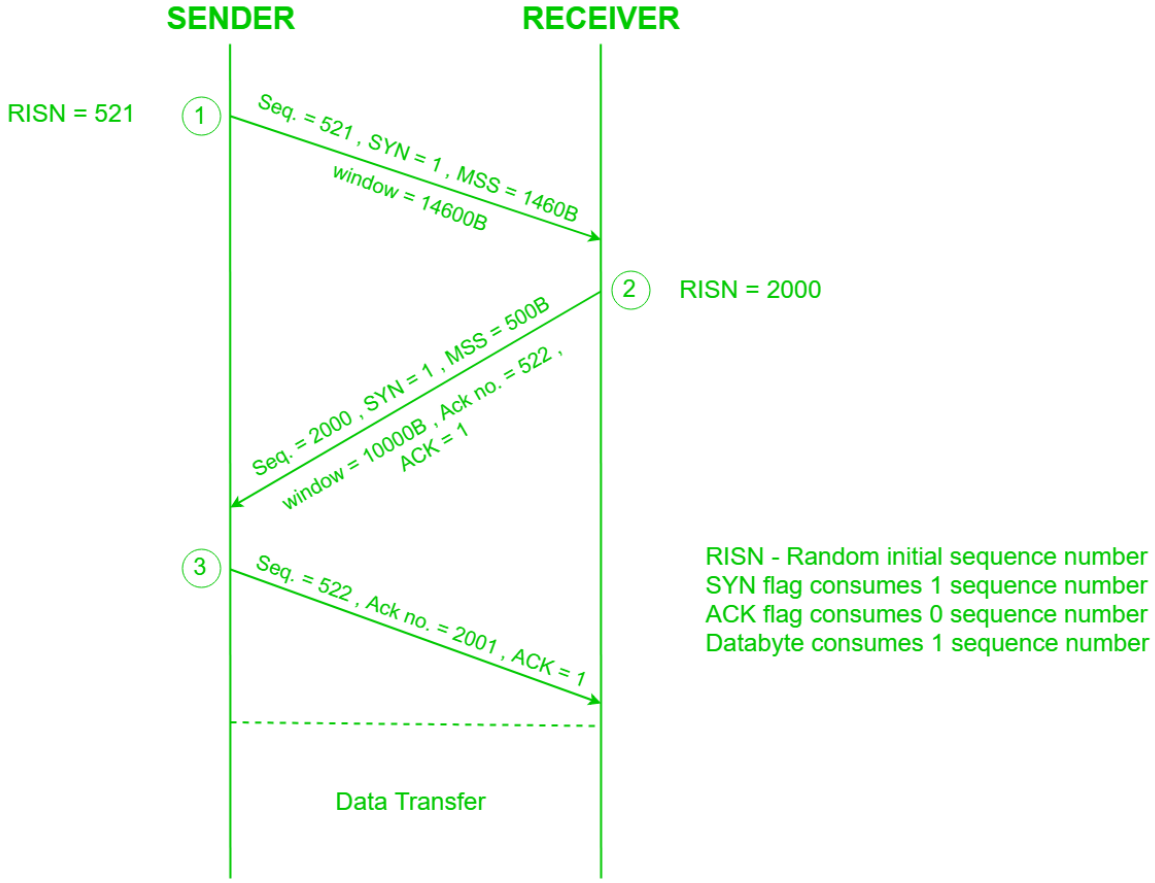
This is the sender's sending window size.

- **Acknowledgement Number (Ack no.=522):** Since sequence number 521 is received by the receiver so, it makes a request for the next sequence number with Ack no.=522 which is the next packet expected by the receiver since Syn flag consumes 1 sequence no.
- **ACK flag (ACk=1):** tells that the acknowledgement number field contains the next sequence expected by the receiver.

3. Sender makes the final reply for connection establishment in the following way:

- **Sequence number (Seq=522):** since sequence number = 521 in 1st step and SYN flag consumes one sequence number hence, the next sequence number will be 522.

- **Acknowledgement Number (Ack no.=2001):** since the sender is acknowledging SYN=1 packet from the receiver with sequence number 2000 so, the next sequence number expected is 2001.
- **ACK flag (ACK=1):** tells that the acknowledgement number field contains the next sequence expected by the sender.



FLOW CONTROL & ERROR CONROL

AT Flow Control at Transport Layer: In communication at the transport layer, we are dealing with four entities: sender process,• sender transport layer, receiver tr The sending process at the application layer is only a producer. It produces message chunks and• pushes them to the transport layer. The sending transport layer has a double role: it is both a consumer and a producer. It consumes the messages pushed by the producer. It encapsulates the messages in packets and• pushes them to the receiving transport layer. The receiving transport layer also has a double role: it is the consumer for the packets received• from the sender and the producer that decapsulates the messages and delivers them to the application layer. Fig: Flow control at the transport layer. Buffers: Although flow control can be implemented in several ways, one of the solutions is normallyto• use two buffers: one at the sending transport layer and the other at the receivingtransport layer. A buffer is a set of memory locations that can hold packets at thesender and receiver.• Error Control: Error control at the transport layer is responsible for• ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY CS8591 COMPUTER NETWORKS FLOW CONTROL ,ERROR CONROL AND CONGESTION CONTROL AT TRANSPORT LAYER In communication at the transport layer, we are dealing with four entities: sender process, sender transport layer, receiver transport layer, and receiver process. The sending process at the application layer is only a producer. It produces message chunks and pushes them to the transport layer. The sending transport layer has a double role: it is both a It consumes the messages pushed by the producer. It encapsulates the messages in packets and pushes them to the receiving transport layer. The receiving transport layer also has a double role: it is the consumer for the packets received d the producer that decapsulates the messages and delivers them to the Flow control at the transport layer. Although flow control can be implemented in several ways, one of the solutions is normallyto ne at the sending transport layer and the other at the receivingtransport layer. A buffer is a set of memory locations that can hold packets at thesender and receiver. Error control at the transport layer is responsible for ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY CS8591 COMPUTER NETWORKS FLOW CONTROL ,ERROR CONROL AND CONGESTION CONTROL AT In communication at the transport layer, we are dealing with four entities: sender process, The sending process at the application layer is only a producer. It produces message chunks and pushes them to the transport layer. The sending transport layer has a double role: it is both a It consumes the messages pushed by the producer. It encapsulates the messages in packets and The receiving transport layer also has a double role: it is the consumer for the packets received d the producer that decapsulates the messages and delivers them to the Although flow control can be implemented in several ways, one of the solutions is normallyto ne at the sending transport layer and the other at the receivingtransport layer. A buffer is a set of memory locations that can hold packets at thesender and receiver. ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY CS8591 COMPUTER NETWORKS 1. Detecting and discarding corrupted packets. 2. Keeping track of lost and discarded packets and resending them. 3. Recognizing duplicate packets and discarding them. 4. Buffering out-of-order packets until the missing packets arrive. Sequence Numbers Error control requires that the sending transport layer knows

which packet is to be resent and• the receiving transport layer knows which packet is a duplicate, or which packet has arrived out of order. This can be done if the packets are numbered. We can add a field to the transport-layer packet to hold the sequence number of the packet.• When a packet is corrupted or lost, the receiving transport layer can somehow inform the sending transport layer to resend that packet using the sequence number. The receiving transport layer can also detect duplicate packets if two received packets have the• same sequence number. The out-of-order packets can be recognized by observing gaps in the sequence numbers.• Acknowledgment: The receiver side can send an acknowledgment (ACK) for each of a collection of packets that• have arrived safe and sound. If an ACK does not arrive before the timer expires, the sender resends the packet.• Combination of Flow and Error Control: Flow control requires the use of two buffers, one at the sender site and the other at the receiver• site. We have also discussed that error control requires the use of sequence and acknowledgment numbers by both sides. These two requirements can be combined if we use two numbered buffers, one at the sender,• one at the receiver. Sliding Window: The buffer is represented as a set of slices, called the sliding window, that occupies part of the• circle at any time. At the sender site, when a packet is sent, the corresponding slice is marked. ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY CS8591 COMPUTER NETWORKS When all the slices are marked, it means that the buffer is full and no further messages can be• accepted from the application layer. When an acknowledgment arrives. Fig: Sliding window in circular format. Most protocols show the sliding window using linear representation. The idea is the same.• Fig: Sliding window in linear format. Congestion Control: ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY CS8591 COMPUTER NETWORKS Important issue in a packet-switched network, such as the Internet, is congestion. Congestion in• a network may occur if the load on the network—the number of packets sent to the network— is greater than the capacity of the network—the number of packets a network can handle. Congestion control refers to the mechanisms and techniques that control the congestion and• keep the load below the capacity. Connectionless and Connection-Oriented Protocols: A transport-layer protocol, like a network-layer protocol, can provide two types of services:• connectionless and connection-oriented. Connectionless Service: In a connectionless service, the source process (application program) needs to divide its• message into chunks of data of the size acceptable by the transport layer and deliver them to the transport layer one by one. The transport layer treats each chunk as a single unit without any relation between the chunks.• When a chunk arrives from the application layer, the transport layer encapsulates it in a packet and sends it. To show the independency of packets, assume that a client process has three chunks of• messages to send to a server process. The packets may arrive out of order at the destination and will be delivered out oforder to the• server process. Fig: Connectionless service: ROHINI COLLEGE OF ENGINEERING AND TECHNOLOGY CS8591 COMPUTER NETWORKS If these three chunks of data belong to the same message, the server process may have received• a strange message. The situation would be worse if one of the packets were lost. Since there is no numbering on the packets, the receiving transport layer has no idea that one of• the messages has been lost. It just delivers two chunks of data to the server

process. Two problems arise from the fact that the two transport layers do not coordinate with each• other. No flow control, error control, or congestion control can be effectively implemented in a• connectionless service.

Error handling in TCP

TCP includes mechanisms for detecting corrupted segments, lost segments, out-of-order segments and duplicated segments.

Error control in TCP is achieved through use of three simple techniques:

- Checksum
- Acknowledgement
- Retransmission

1. Checksum

- Each segment includes a checksum field used to check for corrupted segment.
- TCP uses a 16-bit checksum.
- Corrupted segment is discarded by the destination and is considered lost.
- TCP checksum is a weak check by modern standards which uses CRC method.

2. Acknowledgement

- TCP confirms the receipt of data segments.
- Control segments that carry no data but consume a sequence number are also acknowledged.
- ACK segments are never acknowledged.

3. Retransmission

- A segment is retransmitted on two occasions:

  i. When a retransmission timer expires

  ii. When the sender receives three duplicate ACKs
- There is no retransmission for ACK segments
- Retransmission after RTO

  i. TCP maintains one RTO timer for all outstanding (sent, but not acknowledged) segments.

  ii. When the timer matures, the earliest outstanding segment is retransmitted.

  iii. Value of RTO is dynamic and updated based on RTT.
- Retransmission in TCP can be Fast retransmission or Adaptive retransmission.

Flow control in TCP

TCP uses an end-to-end flow control protocol to avoid having the sender send data too fast for the TCP receiver. It uses sliding window protocol for this.

In each TCP segment, the receiver specifies in the receive window field the amount of additionally received data that it is willing to buffer for the connection.

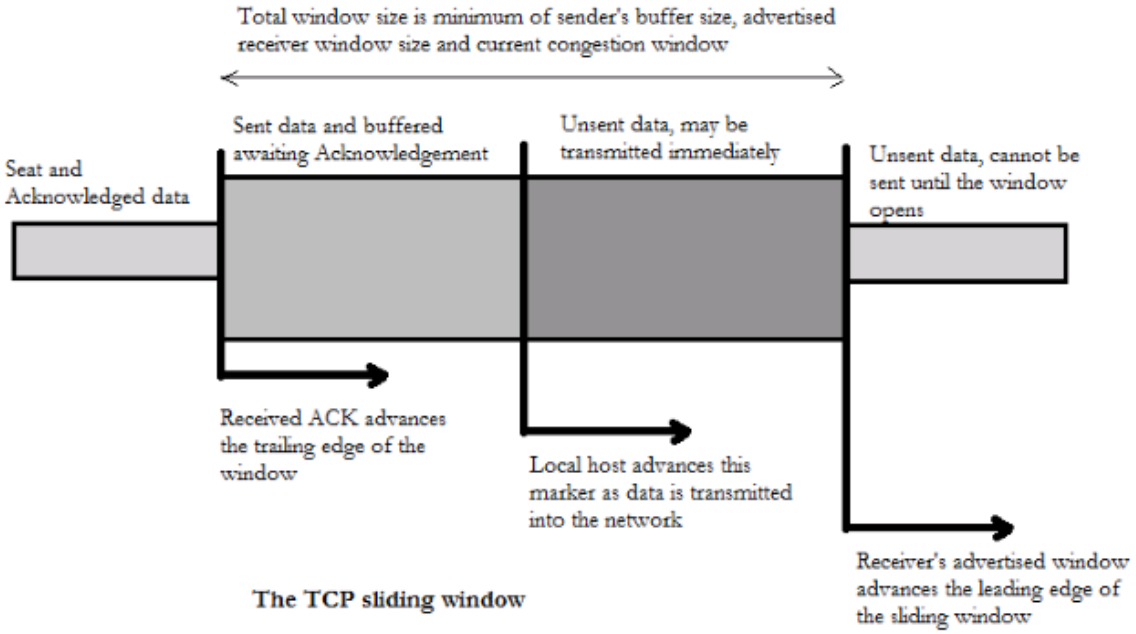
The size of the window is determined by the lesser of two values: rwnd or cwnd

i. rwnd

It is the number of bytes the receiver can accept before its buffer overflows.

ii. cwnd

It is the value determined by the network to avoid congestion.



The TCP sliding window

TCP CONGESTION CONTROL

TCP congestion control is a method used by the TCP protocol to manage data flow over a network and prevent congestion. TCP uses a congestion window and congestion policy that avoids congestion. Previously, we assumed that only the receiver could dictate the sender's window size. We ignored another entity here, the network. If the network cannot deliver the data as fast as it is created by the sender, it must tell the sender to slow down. In other words, in addition to the receiver, the network is a second entity that determines the size of the sender's window

Congestion Policy in TCP

1. **Slow Start Phase:** Starts slow increment is exponential to the threshold.
2. **Congestion Avoidance Phase:** After reaching the threshold increment is by 1.
3. **Congestion Detection Phase:** The sender goes back to the Slow start phase or the Congestion avoidance phase.

Slow Start Phase

Exponential increment: In this phase after every RTT the congestion window size increments exponentially.
Example:- If the initial congestion window size is 1 segment, and the first segment is successfully acknowledged, the congestion window size becomes 2 segments. If the next transmission is also acknowledged, the congestion window size doubles to 4 segments. This exponential growth continues as long as all segments are successfully acknowledged.
Initially cwnd = 1

After 1 RTT, cwnd = 2^(1) = 2

2 RTT, cwnd = 2^(2) = 4

3 RTT, cwnd = 2^(3) = 8

Congestion Avoidance Phase

Additive increment: This phase starts after the threshold value also denoted as ssthresh. The size of cwnd(congestion window) increases additive. After each RTT cwnd = cwnd + 1.
Example:- if the congestion window size is 20 segments and all 20 segments are successfully acknowledged within an RTT, the congestion window size would be increased to 21 segments in the next RTT. If all 21 segments are again successfully acknowledged, the congestion window size would be increased to 22 segments, and so on.
Initially cwnd = i

After 1 RTT, cwnd = i+1

2 RTT, cwnd = i+2

3 RTT, cwnd = i+3

Congestion Detection Phase

Multiplicative decrement: If congestion occurs, the congestion window size is decreased. The only way a sender can guess that congestion has happened is the need to retransmit a segment. Retransmission is needed to recover a missing packet that is assumed to have been

dropped by a router due to congestion. Retransmission can occur in one of two cases: when the RTO timer times out or when three duplicate ACKs are received.

Case 1: Retransmission due to Timeout – In this case, the congestion possibility is high.

(a) ssthresh is reduced to half of the current window size.

(b) set cwnd = 1

(c) start with the slow start phase again.

Case 2: Retransmission due to 3 Acknowledgement Duplicates – The congestion possibility is less.

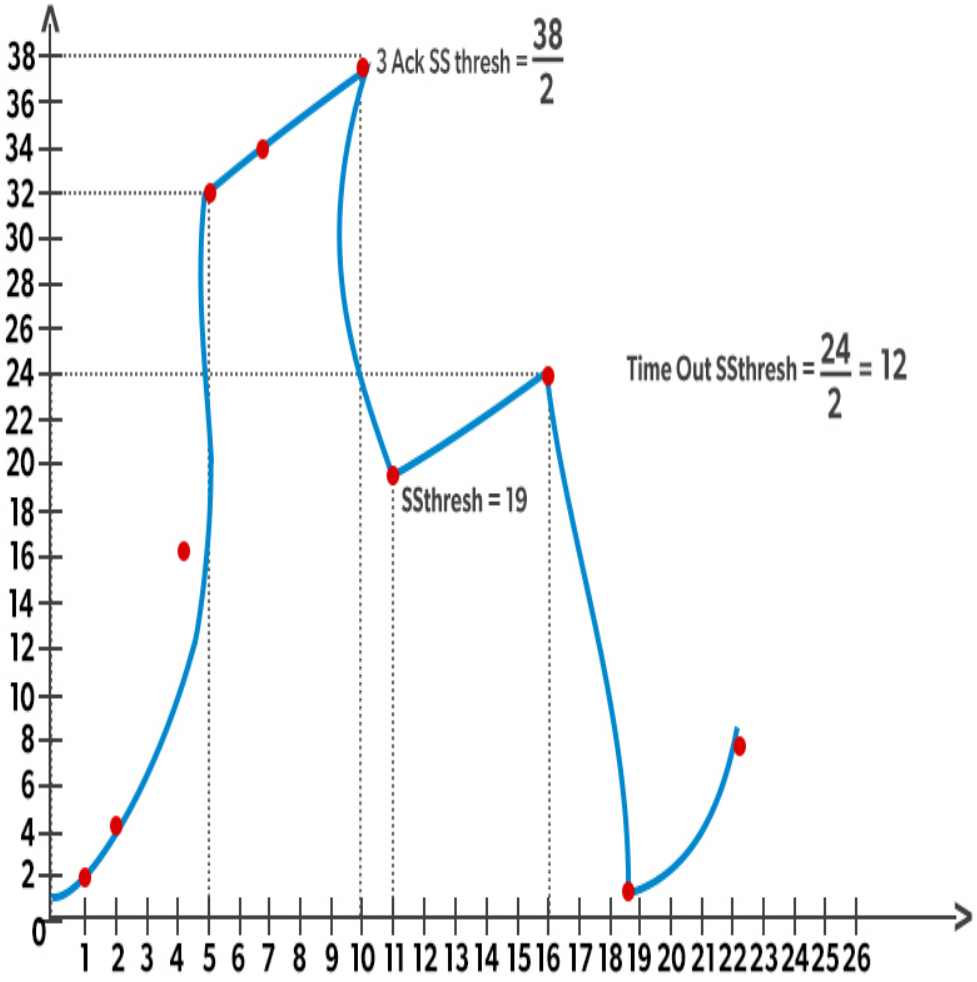
(a) ssthresh value reduces to half of the current window size.

(b) set cwnd= ssthresh

(c) start with congestion avoidance phase

Example

Assume a TCP protocol experiencing the behavior of slow start. At the 5th transmission round with a threshold (ssthresh) value of 32 goes into the congestion avoidance phase and continues till the 10th transmission. At the 10th transmission round, 3 duplicate ACKs are received by the receiver and entered into additive increase mode. Timeout occurs at the 16th transmission round. Plot the transmission round (time) vs congestion window size of TCP segments.



TCP AND UDP IN TRANSPORT LAYER

Layer 3 or the Network layer uses IP or Internet Protocol which being a connection less protocol treats every packet individually and separately leading to lack of reliability during a transmission. For example, when data is sent from one host to another, each packet may take a different path even if it belongs to the same session. This means the packets may/may not arrive in the right order. Therefore, IP relies on the higher layer protocols to provide reliability.

TCP (Transmission Control Protocol):

TCP is a layer 4 protocol which provides acknowledgement of the received packets and is also reliable as it resends the lost packets. It is better than UDP but due to these features it has an additional overhead. It is used by application protocols like HTTP and FTP.

UDP (User Datagram Protocol):

UDP is also a layer 4 protocol but unlike TCP it doesn't provide acknowledgement of the sent packets. Therefore, it isn't reliable and depends on the higher layer protocols for the same. But on the other hand it is simple, scalable and comes with lesser overhead as compared to TCP. It is used in video and voice streaming.

TCP Vs UDP –

1. **Session Multiplexing:**

   A single host with a single IP address is able to communicate with multiple servers. While using TCP, first a connection must be established between the server and the receiver and the connection is closed when the transfer is completed. TCP also maintains reliability while the transfer is taking place.
   UDP on the other hand sends no acknowledgement of receiving the packets. Therefore, provides no reliability.

2. **Segmentation:**
   Information sent is first broken into smaller chunks for transmission.
   Maximum Transmission Unit or MTU of a Fast Ethernet is 1500 bytes whereas the theoretical value of TCP is 65495 bytes. Therefore, data has to be broken into smaller chunks before being sent to the lower layers. MSS or Maximum Segment Size should be set small enough to avoid fragmentation. TCP supports MSS and Path MTU discovery with which the sender and the receiver can automatically determine the maximum transmission capability.

   UDP doesn't support this; therefore it depends on the higher layer protocols for data segmentation.

3. **Flow Control:**

   If sender sends data faster than what receiver can process then the receiver will drop the data and then request for a retransmission, leading to wastage of time and resources. TCP provides end-to-end flow control which is realized using a sliding window. The sliding

window sends an acknowledgement from receiver's end regarding the data that the receiver can receive at a time.
UDP doesn't implement flow control and depends on the higher layer protocols for the same.

4. **Connection Oriented:**

   TCP is connection oriented, i.e., it creates a connection for the transmission to take place, and once the transfer is over that connection is terminated.
   UDP on the other hand is connectionless just like IP (Internet Protocol).

5. **Reliability:**
   TCP sends an acknowledgement when it receives a packet. It requests a retransmission in case a packet is lost.
   UDP relies on the higher layer protocols for the same.

6. **Headers:**

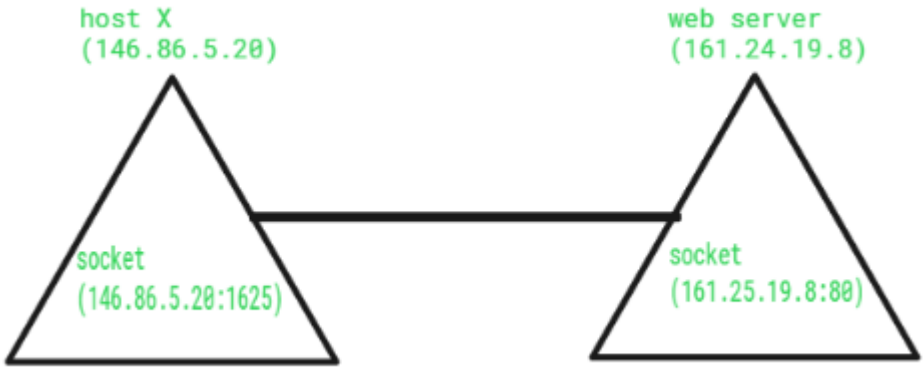
The size of TCP header is 20-bytes (16-bits for source port, 16-bits for the destination port, 32-bits for seq number, 32-bits for ack number, 4-bits header length)

The size of the UDP header is 8-bytes (16-bits for source port, 16-bits for destination port, 16-bits for length, 16-bits for checksum); it's significantly smaller than the TCP header.

Both UDP and TCP header is comprised of 16-bit Source port(these are used for identifying the port number of the source) fields and 16-bits destination port (these are used for specifying the offered application) fields.

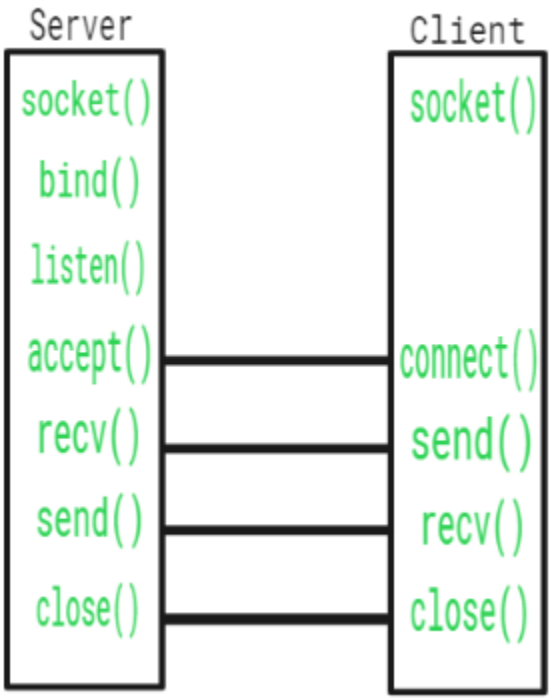
SOCKET IN COMPUTER NETWORK

A **socket** is one endpoint of a **two way** communication link between two programs running on the network. The socket mechanism provides a means of inter-process communication (IPC) by establishing named contact points between which the communication take place.
Like 'Pipe' is used to create pipes and sockets is created using **'socket'** system call. The socket provides bidirectional **FIFO** Communication facility over the network. A socket connecting to the network is created at each end of the communication. Each socket has a specific address. This address is composed of an IP address and a port number.
Socket are generally employed in client server applications. The server creates a socket, attaches it to a network port addresses then waits for the client to contact it. The client creates a socket and then attempts to connect to the server socket. When the connection is established, transfer of data takes place.



Types of Sockets : There are two types of Sockets: the **datagram** socket and the **stream** socket.

1. **Datagram Socket :** This is a type of network which has connection less point for sending and receiving packets. It is similar to mailbox. The letters (data) posted into the box are collected and delivered (transmitted) to a letterbox (receiving socket).
2. **Stream Socket** In Computer operating system, a stream socket is type of interprocess communications socket or network socket which provides a connection-oriented, sequenced, and unique flow of data without record boundaries with well defined mechanisms for creating and destroying connections and for detecting errors. It is similar to phone. A connection is established between the phones (two ends) and a conversation (transfer of data) takes place.



Function Call	**Description**
Socket()	To create a socket
Bind()	It’s a socket identification like a telephone number to contact
Listen()	Ready to receive a connection
Connect()	Ready to act as a sender
Accept()	Confirmation, it is like accepting to receive a call from a sender
Write()	To send data
Read()	To receive data
Close()	To close a connection